IN A COMMUTATIVE TRIANGLE, AFFINENESS OF ONE LEG AND OF THE OTHER DIAGONAL GIVES AFFINENESS OF THE THIRD MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change permanence of affineness. Invoked twice inside the counted proof (source lines 2374-2376 and 2379-2382). Makes the projection of the fibre product affine and the graph morphism affine.
- The identification of the graph with a base change of the diagonal of the second leg. Used inside the counted proof (source lines 2377-2381). Realizes the graph morphism as a base change of that diagonal.
- The composition permanence of affineness. Invoked at the end of the counted proof (source lines 2383-2385). Combines the two affine factors into the assertion.

Source reference. Original reference categories-section-2-fibre-products: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2382–2382. Navigation pointer only: the counted text refers to the Stacks item categories-section-2-fibre-products, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-base-change-affine: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2376–2376. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-affine, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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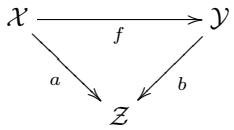
Source reference. Original lemma lemma-composition-affine: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2386–2386. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-affine, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of algebraic stacks. If a is affine and Δ_b is affine, then f is affine.

Proof. The base change $\mathrm{pr}_2 : \mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{Y}$ of a is affine by Lemma lemma-base-change-affine. The morphism $(1, f) : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{Y}$ is the base change of $\Delta_b : \mathcal{Y} \to \mathcal{Y} \times_{\mathcal{Z}} \mathcal{Y}$ by the morphism $\mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{Y} \times_{\mathcal{Z}} \mathcal{Y}$ (see material in Categories, Section categories-section-2-fibre-products). Hence it is affine by Lemma lemma-base-change-affine. The composition $f = \mathrm{pr}_2 \circ (1, f)$ of affine morphisms is affine by Lemma lemma-composition-affine and the proof is done. □